The posterior mean removes the avoidable squared error

Conditional squared-error risk Previous slide: residual uncertainty plus squared distance from the mean

$$R(a|y) = \operatorname{tr} V + \|m - a\|^2$$

Choose a to minimize $R(a|y)$

$$\|m - a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \operatorname{tr} V$$

For the conditional Gaussian: $a^*(y) = \mu_{z|y}$

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.